Joshua Walker

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Education

Imperial College London *(2022 - 2026)*

MEng Computing, fourth year - on track for achieving **first class** honours

Employment History

Elliott Management *(Apr 2025 - Sep 2025)*

Worked as a **Junior Software Engineer** for the London branch of New York-based hedge fund **Elliott Management**, supporting senior developers in their roles - key projects included providing traders with reports to aid market assessments and developing stronger software engineering foundations for the monitoring of support systems.

Technical Skills

Languages and Technologies

Scala, ScalaJS, ScalaNative, Rust, C, C#, Unity, Java, Haskell, Python, JS, ReactJS, HTML, CSS, Assembly, Git, Bash, Latex

Projects

Ginseng *(Final Year Project)* - Scala, OpenGL *(2026)*

Proposed and developed a project utilising principled staged-metaprogramming techniques to efficiently convert from an ergonomic functional programming domain specific language for rendering graphics into standardised OpenGL code.

Dill *(Group Project)* - Rust, Scala, Tauri, Laminar *(2025)*

Built a **interactive cross-platform debugger** for the parser combinator library Parsley (utilised in Imperial's compilers coursework), working throughout semester to deliver an app meeting both the brief of our supervisor and the needs of our fellow undergraduates.

Designing for Real People *(Group Project)* - Flutter, Firebase, Dart *(2024)*

Studied **Human-Centered Design** with the Royal College of Art in order to build a recipe-sharing social media app for students, using user interviews to ensure our product was commercially viable and practical for our users.

Sorts TUI - Rust *(2024)*

Developed an in-terminal utility for visualising sorting algorithms, built as as an exercise in understanding **algorithmic complexity**.

Portfolio Website - JavaScript, React, CSS *(2024)*

Designed and developed my own portfolio website to showcase skills and experience gained through university and solo projects.

WACC *(Group Project)* - Rust *(2024)*

Built an **optimising compiler** in Rust using the Chumsky parser library, compiling initially to an internal intermediate representation to allow for code generation into both Arm32 and Intel x86-64 assembly code.

Pintos *(Group Project)* - C *(2023)*

Extended a simple **operating system** with features such as synchronisation primitives, system calls and virtual memory. Explored foundational concepts of OS architectures and delved into the pre-existing codebase to research their practical implementations.

Weather Wallpaper - Rust *(2022 - 2024)*

Built a command-line **wallpaper engine** that changes a user's desktop wallpaper to reflect local weather, learning the fundamentals of using the REST API to access online data.

Origame *(Game Jam submission)* - Unity, C# *(2022 - 2024)*

Created a 2D puzzle-platformer game for A-Level coursework using the mechanics of folding paper platforms to progress through levels.

Extracurricular

Self-taught guitarist | enjoy competing in Game Jams | studied Practical Art as an optional module | former member of an archery club | volunteered and participated in Europe's largest student-run hackathon, hosted on Imperial's South Kensington Campus.